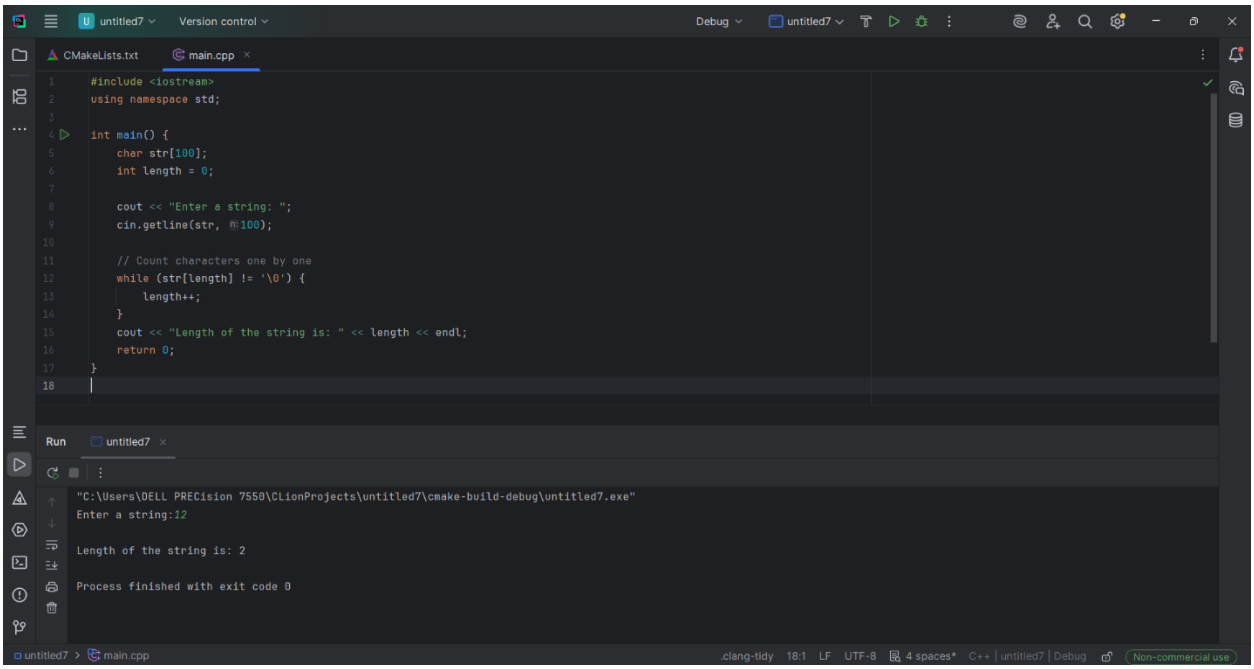
PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Programming Fundamentals Lab 11	**Instructor: Mr. M Hamza**
Program: AI Fall 2025	**Date:15 january, 2026**
Course Code: COMP-111L	

Lab Task

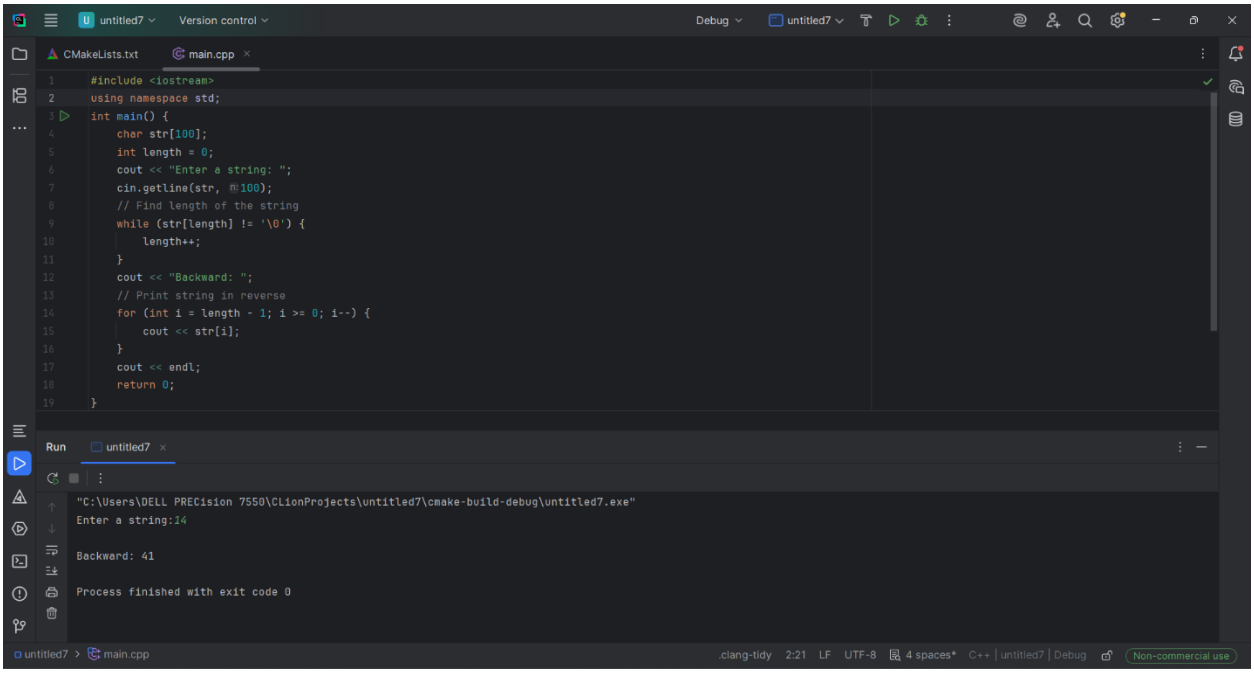
1. String Length

Write a function that returns an integer and accepts a pointer to a C-string as an argument. The function should count the number of characters in the string and return that number. Demonstrate the function in a simple program that asks the user to input a string, passes it to the function, and then displays the function's return value.



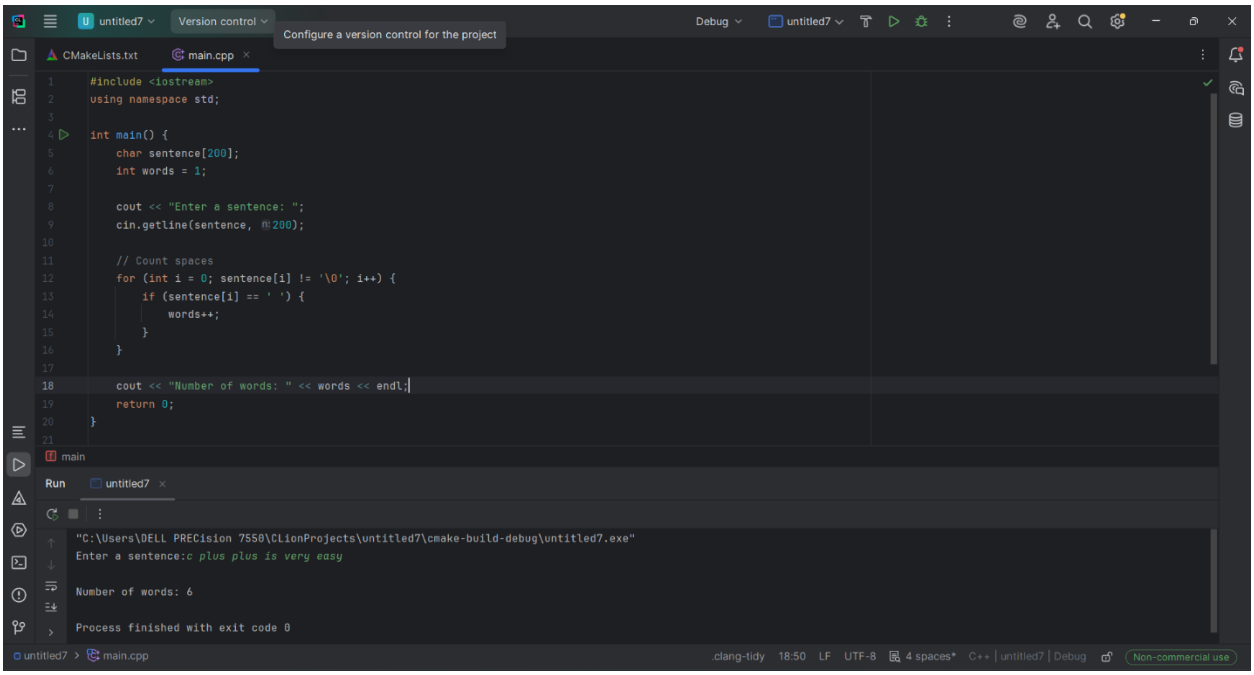
2. Backward String

Write a function that accepts a pointer to a C-string as an argument and displays its contents backward. For instance, if the string argument is "Gravity" the function should display "ytivarG". Demonstrate the function in a program that asks the user to input a string and then passes it to the function.



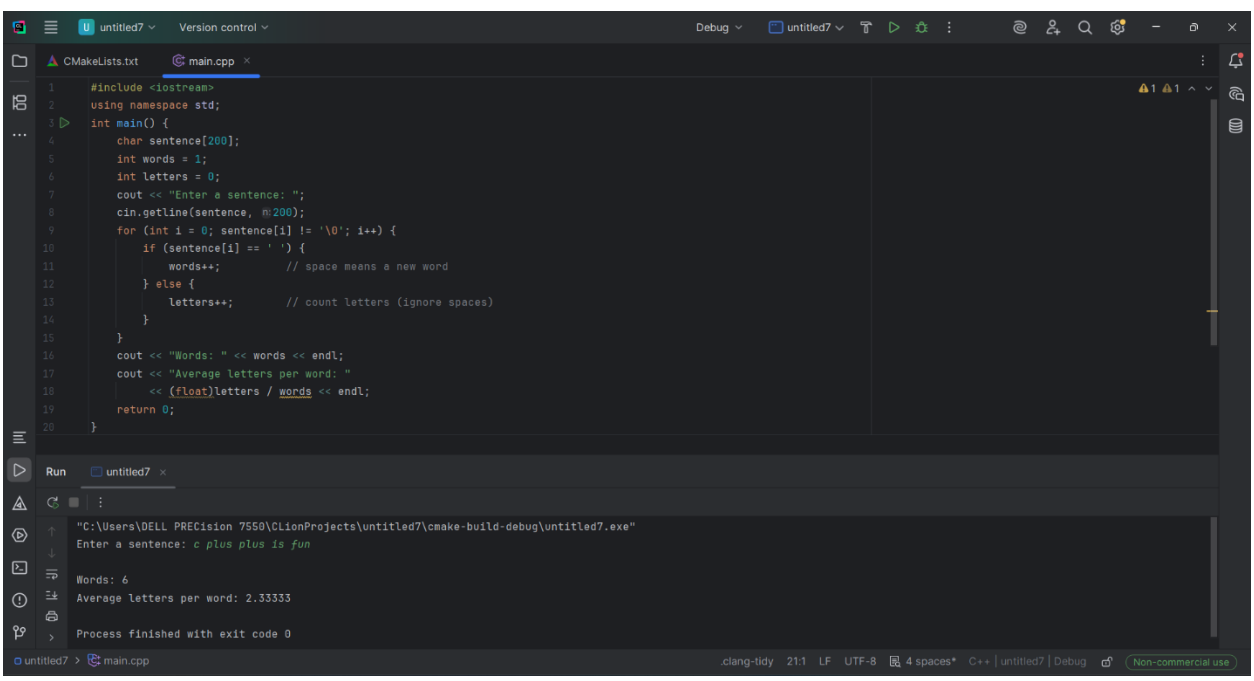
3. Word Counter

Write a function that accepts a pointer to a C-string as an argument and returns the number of words contained in the string. For instance, if the string argument is “Four score and seven years ago” the function should return the number 6. Demonstrate the function in a program that asks the user to input a string and then passes it to the func- tion. The number of words in the string should be displayed on the screen. Optional Exercise: Write an overloaded version of this function that accepts a string class object as its argument.

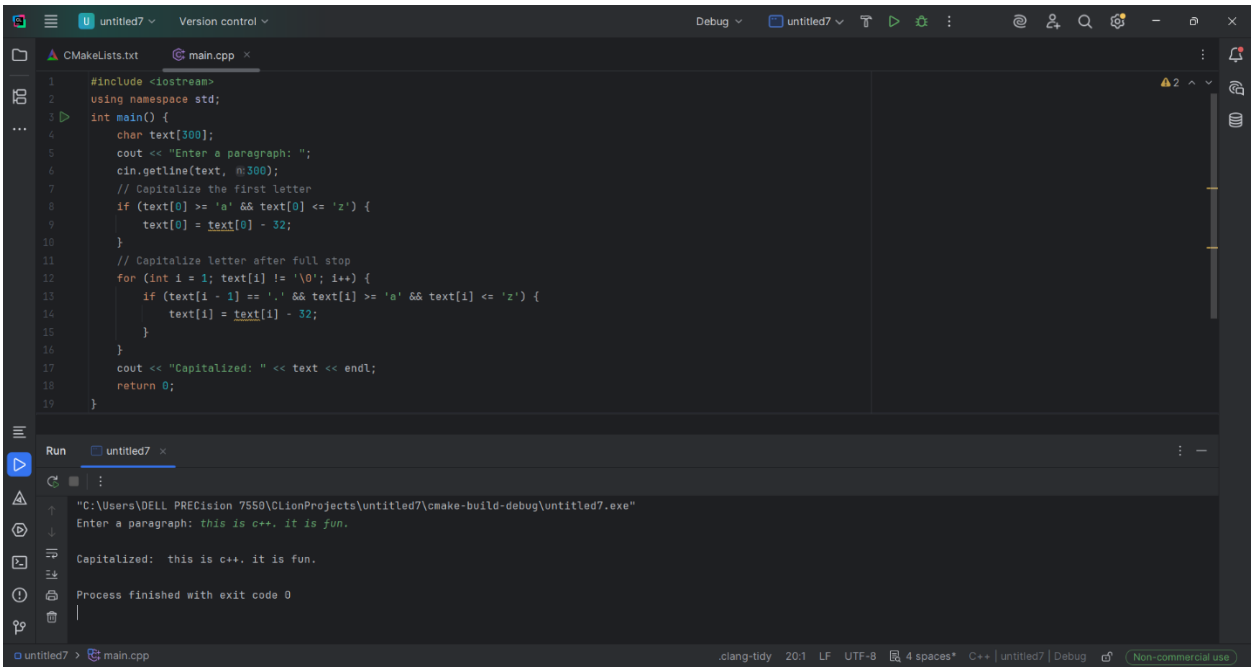


4. Average Number of Letters

Modify the program you wrote for Problem 3 (Word Counter), so it also displays the average number of letters in each word.



5. Sentence Capitalizer

Write a function that accepts a pointer to a C-string as an argument and capitalizes the first character of each sentence in the string. For instance, if the string argument is “hello. my name is

Joe. what is your name?” the function should manipu- late the string so it contains “Hello. My name is Joe. What is your name?”